

Predictors of Statistics Anxiety Among University Students

An Applied Regression Analysis

Yanan He Emily Hershey Kylie Houston Tianai Huang
Xinwan Hu

Purdue University
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Presentation Outline

- ① Background and Research Motivation
- ② Dataset and Key Variables
- ③ Exploratory Data Analysis
- ④ Regression Analyses
 - ANOVA via dummy-variable regression
 - Partial F -test
 - Lack-of-fit test
 - Moderation analysis
 - Model selection and Validation
- ⑤ Key Findings and Conclusions

What predicts students' statistics anxiety?

Statistics anxiety is common among university students because statistics courses are required across many majors.

This project uses regression analysis to examine which psychological and academic factors are associated with different dimensions of statistics anxiety.

Data Source

Open dataset accompanying McCaughey et al.'s study on statistics and math anxiety.
<https://osf.io/nzhq6/>

Main Predictors

Math anxiety, anxiety sensitivity, perfectionism, self-efficacy, faculty, and prior statistics experience.

Analytical Goal

Use applied regression methods to identify key predictors of statistics anxiety.

Why Statistics Anxiety Matters

Background

- Statistics courses are required in many university programs.
- Students may feel anxious during tests, interpretation tasks, or help-seeking situations.
- Statistics anxiety is therefore a common academic challenge among university students.

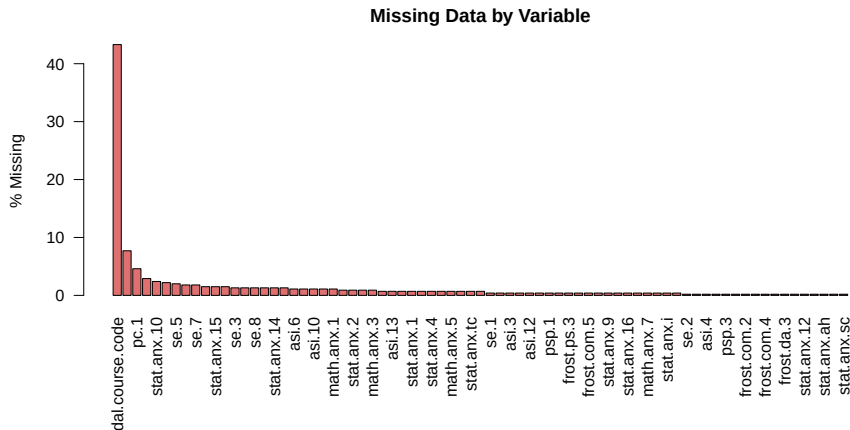
Motivation

- Statistics anxiety may affect students' confidence, performance, and engagement.
- Different students may experience different dimensions of statistics anxiety.
- This project examines which factors are associated with these anxiety dimensions.

Dataset Overview

- Dataset: *The Association of Self-Efficacy, Anxiety Sensitivity, and Perfectionism with Statistics and Math Anxiety*
- Sample: university students
- Working dataset: 453 responses and 97 variables
- Variables include:
 - Likert-scale questionnaire items
 - Composite psychological scale scores
 - Demographic and academic background variables
- Missingness was limited; complete cases were used for each regression model.

Missing Data Check



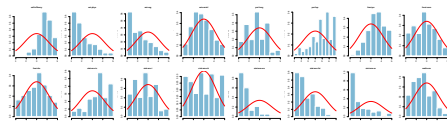
Overall missingness was limited, so the regression analyses used complete cases for the variables required in each model.

Key Variables Used in the Analysis

Role	Variables	Purpose
Outcomes	STARS subscales	Measure statistics anxiety
Math anxiety	math.anx	Main predictor
Anxiety sensitivity	ASI-3 subscales	Incremental prediction
Perfectionism	frost.com	Lack-of-fit analysis
Self-efficacy	self.efficacy	Model selection
Background	Faculty, prior statistics	Group and moderation analyses

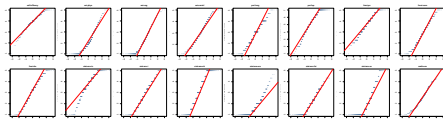
These variables motivate the EDA and regression analyses that follow.

Distribution and Normality Check



Distribution of Composite Scores

- Composite scores show mixed shapes.
- Some anxiety measures are mildly right-skewed.

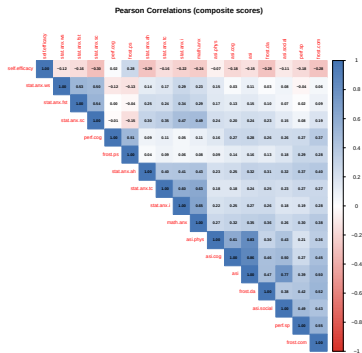


Q-Q Plots of Composite Scores

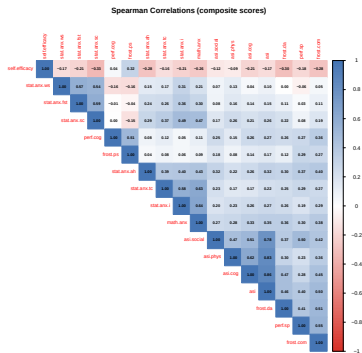
- Most variables are close to normal in the center.
- Several variables show mild tail deviations.

These patterns motivate checking regression assumptions using residual diagnostics later.

Correlation Structure



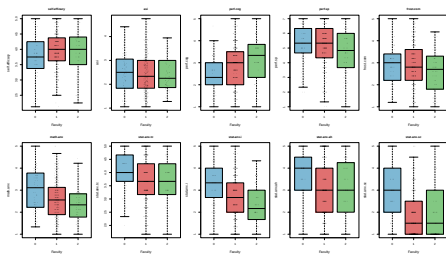
Pearson Correlation
Measures linear relationships.



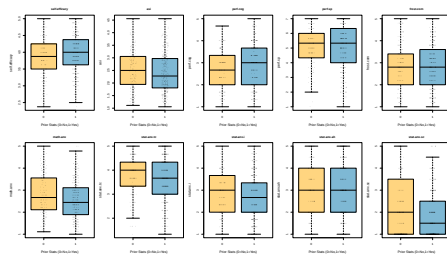
Spearman Correlation
Measures rank-based monotonic relationships.

Both matrices show similar patterns: math anxiety is positively related to several statistics anxiety measures, while self-efficacy tends to be negatively related to anxiety.

Group-Based Descriptive Comparisons



Anxiety outcomes by academic faculty



Anxiety outcomes by prior statistics experience

Takeaway: The boxplots suggest possible group differences by faculty and prior statistics experience, but the distributions also overlap substantially. These patterns motivate the later ANOVA and moderation analyses.

RQ1:ANOVA via Dummy-Variable Regression

Research Question

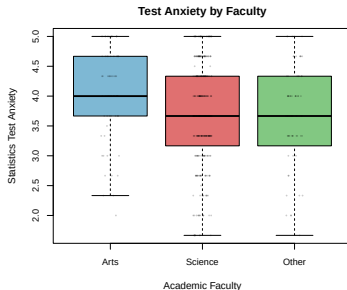
Does statistics test anxiety differ across academic faculties?

Using Arts as the reference group, we fit the dummy-variable regression model:

$$\hat{Y}_i = 3.978 - 0.345D_{\text{Science},i} - 0.434D_{\text{Other},i}.$$

Faculty	<i>n</i>	Mean	SD
Arts	67	3.978	0.924
Science	315	3.632	0.948
Other	68	3.544	1.038

Descriptive statistics for statistics test anxiety.



Statistics test anxiety by academic faculty.

Arts students reported the highest average test anxiety, but the group distributions overlap substantially.

RQ1:ANOVA Results and Interpretation

Overall ANOVA Test

$$F(2, 447) = 4.278, \quad p = .014$$

Mean statistics test anxiety differs across faculty groups.

Post-hoc Comparisons

- Arts vs. Science: $p = .021$
- Arts vs. Other: $p = .024$
- Science vs. Other: $p = .771$

Model Fit

$$R^2 = .019$$

Faculty explains only a small share of variation in test anxiety.

Conclusion

The main difference is between Arts students and the two non-Arts groups. The effect is statistically significant but practically modest.

This motivates moving beyond broad faculty groups to psychological predictors such as math anxiety.

RQ2: Incremental Value of Anxiety Sensitivity

Research Question

Does anxiety sensitivity improve the prediction of statistics test anxiety beyond math anxiety alone?

Motivation

Math anxiety is expected to be strongly related to statistics anxiety. This analysis asks whether anxiety sensitivity explains additional variation after math anxiety is already included.

Nested Models

Reduced model: $Y_i = \beta_0 + \beta_1(\text{math.anx})_i + \epsilon_i$

Full model: $Y_i = \beta_0 + \beta_1(\text{math.anx})_i + \beta_2(\text{asi.phys})_i + \beta_3(\text{asi.cog})_i + \beta_4(\text{asi.social})_i$

Partial F-Test

$H_0 : \beta_2 = \beta_3 = \beta_4 = 0$ vs. $H_a : \text{at least one ASI coefficient is nonzero.}$

If the full model fits significantly better, ASI adds predictive value beyond math anxiety.

RQ2: Partial F -Test Result

Goal: Test whether the ASI dimensions jointly improve prediction beyond math anxiety.

Model	Predictors	SSE	R^2	Adj. R^2
Reduced	Math anxiety only	254.25	0.3914	0.3901
Full	Math anxiety + ASI dimensions	249.88	0.4019	0.3965

$$\Delta R^2 = 0.0104 \quad F(3, 444) = 2.5854 \quad p = 0.05268$$

Interpretation: Adding the ASI dimensions explained only about 1.04% additional variation. Since $p = .05268 > .05$, we fail to reject H_0 .

Conclusion: the three ASI dimensions do not jointly add clear predictive value beyond math anxiety.

RQ2: Full Model Coefficient Summary

Full model:

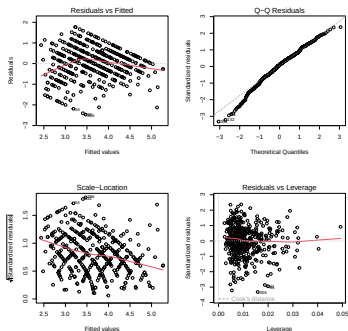
$$Y_i = \beta_0 + \beta_1(\text{math.anx})_i + \beta_2(\text{asi.phys})_i + \beta_3(\text{asi.cog})_i + \beta_4(\text{asi.social})_i + \epsilon_i.$$

Predictor	Estimate	SE	p	Interpretation
math.anx	0.690	0.044	< .001	Strong positive predictor
asi.phys	0.004	0.048	.929	Not significant
asi.cog	-0.073	0.047	.122	Not significant
asi.social	0.121	0.045	.008	Significant positive predictor

Key interpretation: Math anxiety remained the dominant predictor. Among the ASI dimensions, only social anxiety sensitivity had a significant unique association with statistics test anxiety after adjusting for math anxiety and the other ASI dimensions.

Main takeaway: the ASI block was not jointly significant, but asi.social may have a modest unique association.

RQ2: Diagnostics and Robustness



Diagnostic plots for the full RQ2 model.

Diagnostics

- Mild non-constant variance in residual plots.
- Breusch–Pagan: $BP = 26.887$, $p < .001$.
- Q–Q plot showed some tail deviation.
- VIF values were low.
- No dominant influential observation.

Robustness

- HC3 robust standard errors were used.
- Conclusions did not change.
- `math.anx` and `asi.social` remained significant.
- `asi.phys` and `asi.cog` remained non-significant.

Takeaway

The model had some heteroscedasticity, but the main inference was robust.

RQ3: Lack-of-Fit Test

Research Question

Does a first-order linear model adequately describe the relationship between perfectionism-related concern over mistakes and statistics interpretation anxiety?

Motivation

Concern over mistakes may be associated with stronger anxiety when students interpret statistical results. Before interpreting a simple linear slope, we test whether the linear form is adequate or whether a more flexible model is needed.

Models Compared

Linear model: $Y_i = \beta_0 + \beta_1(\text{frost.com})_i + \epsilon_i$

Saturated model: $Y_i = \mu_j + \epsilon_i$ for each observed value/group of `frost.com`

Lack-of-Fit Test

$$H_0 : E(Y | X) = \beta_0 + \beta_1 X \quad \text{vs.} \quad H_a : E(Y | X) \neq \beta_0 + \beta_1 X.$$

If the saturated model fits significantly better, the simple linear model may lack fit.

RQ3: Lack-of-Fit Test Results

Question Tested

Does the saturated model provide a significantly better fit than the first-order linear model?

Model	Error df	SSE	AIC	BIC
Linear	449	413.66	1246.91	1259.24
Saturated	427	387.97	1261.99	1364.77

$$F(22, 427) = 1.2854, \quad p = 0.1751$$

Interpretation

Since $p = .1751 > .05$, we fail to reject H_0 . There is not sufficient evidence that the first-order linear model is inadequate.

Main takeaway: the simpler linear model is adequate for this relationship.

RQ3: Linear Model Interpretation

Selected Model

Because the lack-of-fit test did not reject linearity, we interpret the first-order linear model.

Predictor	Estimate	SE	<i>t</i>	<i>p</i>
Intercept	1.920	0.139	13.80	< .001
frost.com	0.287	0.046	6.22	< .001

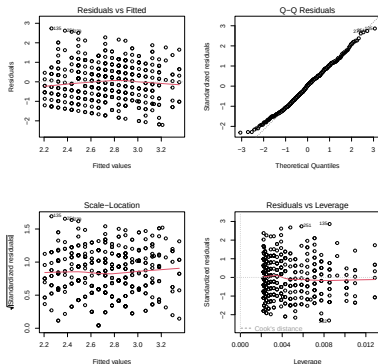
$$R^2 = 0.0793, \quad \text{Adjusted } R^2 = 0.0772$$

Interpretation

Higher concern over mistakes was associated with higher statistics interpretation anxiety. A one-unit increase in `frost.com` was associated with an estimated 0.287-point increase in interpretation anxiety.

The association is statistically significant, but the model explains a modest amount of variation.

RQ3: Diagnostics



Diagnostic plots for the RQ3 linear model.

Diagnostic Summary

- Residual-versus-fitted plot showed no strong systematic curvature.
- Scale-location plot did not suggest severe non-constant variance.
- Breusch-Pagan test:

$$BP = 0.0381, \quad p = 0.8453.$$

- Q-Q plot was approximately linear with mild tail deviation.
- Shapiro-Wilk test was borderline but not significant:

$$W = 0.9936, \quad p = 0.0550.$$

Diagnostics supported retaining the first-order linear model.

RQ4: Moderation by Prior Statistics Experience

Research Question

Does the association between math anxiety and statistics test anxiety depend on whether a student has prior statistics course experience?

Motivation

Prior statistics experience may change how strongly math anxiety is connected to statistics test anxiety. Students with previous exposure to statistics may connect math-related anxiety more directly to statistics tasks.

Interaction Model

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 (X_i D_i) + \epsilon_i,$$

where $Y_i = \text{stat.anx.tc}$, $D_i = \text{stats.history}$, and

$$X_i = \text{math.anx}_i - \overline{\text{math.anx}}$$

is centered math anxiety.

Moderation Test

$$H_0 : \beta_3 = 0 \quad \text{vs.} \quad H_a : \beta_3 \neq 0.$$

If the interaction is significant, the math-anxiety slope differs by prior statistics experience.

RQ4: Interaction Model Results

Predictor	Estimate	SE	<i>t</i>	<i>p</i>
Intercept	3.720	0.0649	57.3	< .001
math_anx_c	0.545	0.0741	7.36	< .001
stats.history1	-0.055	0.0777	-0.71	.477
Interaction	0.215	0.0891	2.42	.016

Note: math anxiety was centered, so stats.history1 represents the prior-experience group difference at average math anxiety.

$$F(1, 445) = 5.8442, \quad p = .0160$$

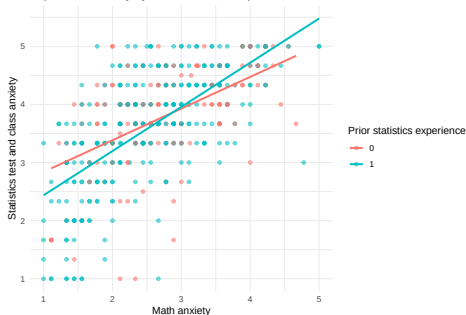
Interpretation

The interaction term was positive and significant, so we reject H_0 . Prior statistics experience significantly moderated the relationship between math anxiety and statistics test anxiety.

Main takeaway: the math anxiety–statistics anxiety association differs by prior statistics experience.

RQ4: Interaction Plot

RQ4: Math Anxiety by Prior Statistics Experience Interaction



Interaction between math anxiety and prior statistics experience.

Slope interpretation

For students without prior statistics experience:

$$\hat{\beta}_{\text{math}} = 0.545.$$

For students with prior statistics experience:

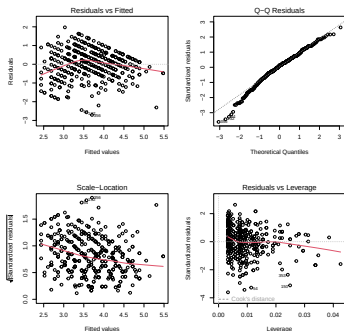
$$\hat{\beta}_{\text{math}} = 0.545 + 0.215 = 0.760.$$

Meaning

Math anxiety was positively associated with statistics test anxiety in both groups, but the association was stronger among students with prior statistics experience.

The positive interaction means the fitted slope is steeper for the prior-experience group.

RQ4: Diagnostics and Robustness



Diagnostic plots for the RQ4 interaction model.

Diagnostics

- Residual plots suggested non-constant residual spread.
- Breusch–Pagan: $BP = 16.864$, $p = .000754$.
- Q–Q plot showed tail deviation.
- VIF values were not severe:

3.28, 1.02, 3.25.

Robustness

- HC3 robust standard errors were examined.
- The interaction remained significant:

$$\hat{\beta}_3 = 0.215, \quad p = .0190.$$

- Some observations were flagged, but no single case was removed.

Takeaway

The model showed heteroscedasticity and tail deviation, but the moderation conclusion was robust.

RQ5: Model Selection for Asking-for-Help Anxiety

Research Question

Which combination of self-efficacy, anxiety sensitivity, and math anxiety best predicts statistics asking-for-help anxiety?

Motivation

Asking-for-help anxiety may be related to different psychological factors: students' confidence in their own ability, broader anxiety sensitivity, and math-specific anxiety. We compared all possible predictor combinations to identify a model that balances fit, parsimony, interpretability, and validation performance.

Candidate Predictors

$$Y_i = \text{stat.anx.ah}$$

Candidate predictors: `self.efficacy`, `asi`, `math.anx`

All $2^3 - 1 = 7$ non-empty subset models were compared.

Selection Criteria

Smaller AIC, BIC, PRESS, LOOCV RMSE, and 10-fold CV RMSE indicate better models; larger adjusted R^2 indicates better penalized explanatory fit.

We did not select based on ordinary R^2 alone because R^2 always increases when predictors are added.

RQ5: Model Selection Results

Model	# Pred.	Adj. R^2	AIC	BIC	10-fold RMSE
SE_ASI_MA	3	0.2458	1361.99	1382.55	1.0916
SE_MA	2	0.2199	1376.25	1392.70	1.1087
ASI_MA	2	0.2160	1378.48	1394.93	1.1128
MA	1	0.1855	1394.71	1407.04	1.1322
SE_ASI	2	0.1567	1411.35	1427.79	1.1542
ASI	1	0.0993	1440.09	1452.42	1.1920
SE	1	0.0812	1449.04	1461.38	1.2043

Selected Model

The full three-predictor model, SE_ASI_MA, was selected because it had the lowest AIC, lowest BIC, highest adjusted R^2 , and lowest 10-fold CV RMSE.

Although this was the most complex candidate model, the criteria consistently favored it. The improvement over the top two-predictor models was modest, but consistent across both information criteria and validation metrics.

Main takeaway: all three predictors contribute useful information for predicting asking-for-help anxiety.

RQ5: Selected Model Coefficients

Selected model

$$\widehat{\text{stat.anx.ah}} = \hat{\beta}_0 + \hat{\beta}_1(\text{self.efficacy}) + \hat{\beta}_2(\text{asi}) + \hat{\beta}_3(\text{math.anx})$$

Predictor	Estimate	SE	p	Interpretation
self.efficacy	-0.361	0.0835	< .001	Negative association
asi	0.280	0.0692	< .001	Positive association
math.anx	0.469	0.0639	< .001	Positive association

$$R^2 = 0.2508, \quad \text{Adjusted } R^2 = 0.2458$$

Interpretation

Holding the other predictors constant, higher self-efficacy was associated with lower asking-for-help anxiety, while higher anxiety sensitivity and math anxiety were associated with higher asking-for-help anxiety.

The selected model explains about 25.1% of the variation in asking-for-help anxiety.

RQ5: Validation and Generalization

Validation Goal

We checked whether the selected model performed well beyond in-sample fit using PRESS, LOOCV RMSE, and 10-fold cross-validation RMSE.

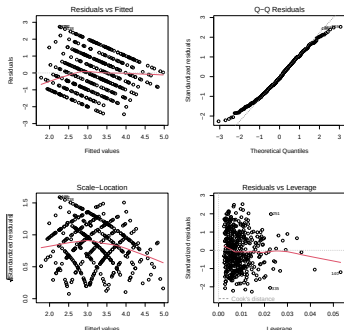
Model	PRESS	LOOCV RMSE	10-fold RMSE
SE_ASI_MA	538.25	1.0925	1.0916
SE_MA	555.77	1.1101	1.1087
ASI_MA	558.25	1.1126	1.1128
MA	578.93	1.1330	1.1322

Validation Interpretation

The selected SE_ASI_MA model had the lowest PRESS, lowest LOOCV RMSE, and lowest 10-fold CV RMSE. Thus, it was not chosen only because it fit the current sample well; it also had the best estimated out-of-sample predictive performance among the candidate models.

Generalization is still limited because the data are observational, cross-sectional, and from one student sample.

RQ5: Diagnostics and Robustness



Diagnostic plots for the selected RQ5 model.

Diagnostics

- Residual plots suggested some non-constant spread.
- Breusch–Pagan: $BP = 11.471$, $p = .00943$.
- Shapiro–Wilk rejected normality:

$$W = 0.9886, \quad p = .00133.$$

- VIF values were low: 1.067, 1.143, and 1.186.
- No single observation dominated the model.

Robustness

- HC3 robust standard errors were used.
- All three predictors remained statistically significant.

Takeaway

Despite some heteroscedasticity and tail deviation, the selected model's substantive conclusions were stable.

Main Statistical Pattern

Statistics anxiety appears to be multidimensional: different predictors are associated with different dimensions of anxiety.

- **Math anxiety** was the most consistent predictor, especially for statistics test and class anxiety.
- **Faculty** differences were statistically significant, but explained only a small amount of variation.
- **Concern over mistakes** had a positive linear association with interpretation anxiety.
- **Prior statistics experience** changed the strength of the math anxiety–statistics anxiety relationship.
- **Asking-for-help anxiety** was best predicted by a combination of self-efficacy, anxiety sensitivity, and math anxiety.

The results suggest that no single factor fully explains statistics anxiety.

Implication for Student Support

Interventions for statistics anxiety may need to address both quantitative anxiety and broader psychological factors.

- Since math anxiety was consistently important, statistics support should include help with mathematical confidence and quantitative reasoning.
- Since self-efficacy predicted asking-for-help anxiety, support programs may benefit from building students' confidence and reducing fear around seeking help.
- Since social anxiety sensitivity showed some unique association in RQ2, some students may need support around embarrassment, visibility, or social evaluation in statistics settings.
- Since prior statistics experience moderated the math anxiety relationship, students with previous exposure may not always be less anxious; their prior experiences may shape how math anxiety connects to statistics anxiety.

*A **targeted support strategy** may be more effective than treating statistics anxiety as one single problem.*

Limitations

- **Observational and cross-sectional data:** the regression models describe associations and prediction patterns, not causal effects.
- **Composite scale outcomes:** the response variables were treated as approximately continuous, but they are still bounded scale-based measures.
- **Assumption concerns:** some models showed heteroscedasticity or mild tail deviations, although robust standard error checks generally preserved the conclusions.
- **Sample-specific results:** the data came from one student sample, so broader generalization requires additional validation.
- **Limited predictors:** course-level factors such as instructor support, grading structure, course format, or prior grades were not included.

*These limitations mean the results should be interpreted as **evidence of association**, not proof of causation.*

How this analysis could be improved

Future work could expand both the data structure and the modeling framework.

- Validate the selected models using an independent student sample.
- Use longitudinal data to examine whether math anxiety, self-efficacy, or anxiety sensitivity predict later changes in statistics anxiety.
- Include instructional variables such as course format, instructor support, assessment style, and prior grades.
- Consider intervention studies to test whether improving self-efficacy or reducing math anxiety lowers statistics anxiety.
- Explore alternative models for bounded scale outcomes if residual diagnostics show strong assumption violations.

A possible next step is to test whether these predictors remain useful across different courses, institutions, and student populations.

Statistics anxiety is not explained by one factor.

Across the models, **math anxiety** was the most consistent predictor, but different dimensions of statistics anxiety were also related to faculty, perfectionism, prior statistics experience, self-efficacy, and anxiety sensitivity.

The strongest practical message is that support for statistics anxiety should address both **quantitative confidence** and broader **psychological barriers**, especially help-seeking concerns.

Applied regression helped identify dimension-specific predictors that may inform more targeted student support.

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